SG98 Remote / Controller Connection Matrix

Final live regression run: 2026-04-26 PDT, generic remote + controller on the bench ports.

Primary rule: ESP-NOW stays available in every normal connectivity state. Wi-Fi enabled adds AP/web access beside ESP-NOW; Wi-Fi disabled parks in ESP-NOW-only. Only

Area	Final Rule	Recovery Meaning
Remote	ESP-NOW first	Factory/no-info remote scans and locks a controller without requiring saved Wi-Fi.
Controller default	Wi-Fi disabled	Default internal state is ESP-NOW-only; Wi-Fi/AP is not a parked state.
Wi-Fi enabled	WEB+ESP	Controller AP/web UI is available while ESP-NOW stays online for the remote.
Full-size only	Radio off allowed	A standalone controller can choose Radio off because the local UI can recover it.
Headless	Radio off rejected	Headless also rejects Wi-Fi-only, then opens WEB+ESP recovery if ESP-NOW traffic is missing.

Build / Flash	Result	Notes
remote-universal-s3	PASS + flashed	Loaded to /dev/cu.usbmodem21101 after Wi-Fi enabled/disabled wording change.
controller-universal-s3	PASS + flashed	Loaded to /dev/cu.usbmodem21201 before and after headless-sim testing; hardware is left on this image.
controller-universal-s3-headless-sim	PASS + flashed/tested	Used only for live headless recovery verification, then replaced by full-size firmware.
dual_board_regression.py	PASS	Both full-size and headless-sim matrices passed from factory/no-info reset.

Operation And Recovery Matrix

Expected paths for full-size, headless, Wi-Fi setup, and remote loss.

Case	Start	Expected Operation	Recovery / Exit
Fresh pair	Both ESP-NOW	Remote scans, discovers controller, locks peer, and polls over ESP-NOW. No saved Wi-Fi required.	Final state is ESP-NOW-only with Wi-Fi/AP off.
Wi-Fi enabled	WEB+ESP	AP/home web UI is available; remote still controls over ESP-NOW.	Controller closes back to ESP-NOW after quiet web activity and healthy ESP-NOW.
Full-size only	ESP-NOW	Controller stays locally usable with no remote present.	User can enable Wi-Fi or choose Radio off locally.
Headless only	ESP-NOW	Controller waits for ESP-NOW traffic.	If silent for the timeout, open WEB+ESP recovery AP.
Remote lost	Was healthy	Headless controller tracks command/status traffic as activity.	If traffic stays missing, stop unsafe running states first, then open WEB+ESP recovery.
Wi-Fi-only request	WEB requested	Full-size diagnostic/bench state only.	Headless rejects it because ESP-NOW would be unavailable.
Radio off	OFF requested	Allowed only for full-size standalone use.	Headless rejects/sanitizes it so the controller cannot strand itself.

Full-Size Live Results

Normal controller build with screen, buttons, encoder, and local recovery enabled.

Case	Remote Avg/Max	Controller Avg/Max	Result	Observed Behavior
Factory/no-info reset	-	-	PASS	Remote network reset + peer clear, controller network reset, no saved Wi-Fi required.
Fresh ESP-NOW discovery	7.8 / 11.2 ms	103.8 / 247.1 ms	PASS	Remote locked controller over ESP-NOW; remote Wi-Fi inactive.
Wi-Fi enabled bridge	14.4 / 41.3 ms	92.6 / 148.9 ms	PASS	Controller stayed WEB+ESP while AP/web access was active; remote stayed ESP-NOW.
Controller-only local use	-	-	PASS	With remote ESP-NOW silent, full-size controller remained locally usable.
Radio off full-size	-	-	PASS	OFF was allowed because local controls can recover the controller.
ESP-NOW parked	12.2 / 68.0 ms	97.6 / 224.2 ms	PASS	Explicit Wi-Fi disabled returned to ESP-NOW-only and remote Wi-Fi stayed off.
Full summary	count 135, avg 8.1, p95 21.4, max 68.0 ms	count 105, avg 148.0, p95 335.6, max 462.3 ms	PASS	ALL_DUAL_BOARD_REGRESSION_PASS

Headless-Sim Live Results

Same physical controller with local UI capabilities disabled for recovery testing.

Case	Remote Avg/Max	Controller Avg/Max	Result	Observed Behavior
Factory/no-info reset	-	-	PASS	Headless-sim started from no saved Wi-Fi and no locked peer.
Fresh ESP-NOW discovery	8.6 / 11.7 ms	87.5 / 96.7 ms	PASS	Remote found and locked the controller without Wi-Fi credentials.
Wi-Fi enabled bridge	7.9 / 10.8 ms	83.3 / 101.9 ms	PASS	WEB+ESP kept ESP-NOW online while recovery/web access was available.
ESP-NOW fallback	7.5 / 11.2 ms	96.1 / 98.9 ms	PASS	When remote ESP-NOW traffic went silent, controller opened WEB+ESP recovery.
Headless guard	9.6 / 24.5 ms	86.7 / 101.0 ms	PASS	OFF and Wi-Fi-only were rejected so the headless controller stayed reachable.
ESP-NOW parked	8.4 / 13.8 ms	89.1 / 97.6 ms	PASS	Returned remote relocked over ESP-NOW; parked state remained low-resource.
Headless summary	count 163, avg 12.9, p95 57.5, max 119.1 ms	count 167, avg 108.6, p95 174.2, max 278.3 ms	PASS	ALL_DUAL_BOARD_REGRESSION_PASS

Corner Cases Covered

What was specifically checked so a controller is not stranded.

Risk	Tested Path	Resolution
Factory-fresh with no Wi-Fi	Remote/controller reset network; WebSocket suite intentionally skipped.	Remote can recover through ESP-NOW discovery without any saved SSID.
Headless boots without remote	Remote transport forced away from ESP-NOW until timeout.	Controller opens WEB+ESP recovery AP.
Headless user selects Radio off	API/menu radio OFF request.	Request rejected with remoteRecoveryRequired=true.
Headless user selects Wi-Fi only	API/menu WEB-only request.	Request rejected; WEB+ESP is the supported recovery bridge.
Full-size standalone owner	Remote ESP-NOW disabled while controller is local.	Controller does not force AP recovery because local screen/buttons/encoder exist.
Wi-Fi/web UI active	WEB+ESP bridge with AP/web activity.	Bridge remains open while active, then can park back to ESP-NOW when quiet and link is healthy.

Report artifact: Documentation/remote-controller-connection-matrix-report.pdf

Source matrix: Documentation/remote-controller-connection-matrix.md